COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	07/15/26	**DATE SUBMITTED:**	07/15/26
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

What procedures did you perform?

Del Mundo.cpp

```cpp
#include <iostream>
using namespace std;

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    const bool validateTriangle();

    double area();
    double perimeter();
    void triangleType();
};

Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

const bool Triangle::validateTriangle() {
    return (totalAngle == 180);
}

double Triangle::area() {
    return (5 * 4) / 2;
}

double Triangle::perimeter() {
    return 5 + 6 + 7;
}

void Triangle::triangleType() {
    if (angleA < 90 && angleB < 90 && angleC < 90)
        cout << "Triangle is Acute-angled.";
    else if (angleA > 90 || angleB > 90 || angleC > 90)
        cout << "Triangle is Obtuse-angled.";
    else
        cout << "Triangle is Others.";
```

```
}

int main() {
    Triangle set1(30, 60, 90);

    if (set1.validateTriangle()) {
        cout << "The shape is a valid triangle." << endl;
        cout << "Area: " << set1.area() << endl;
        cout << "Perimeter: " << set1.perimeter() << endl;
        set1.triangleType();
    }
    else {
        cout << "The shape is NOT a valid triangle.";
    }

    return 0;
}
```

```
The shape is a valid triangle.
Area: 10
Perimeter: 18
Triangle is Others.
--------------------------------
Process exited after 0.04688 seconds with return value 0
Press any key to continue . . .
```

Del Mundo.cpp

```cpp
#include <iostream>
using namespace std;

void sum(int a, int b);
int greaterNum(int A, int B);
bool logicalOperation(bool a, bool b);

int main() {
    int choice;

    cout << "MENU" << endl;
    cout << "1. Sum of Two Numbers" << endl;
    cout << "2. Greater Number" << endl;
    cout << "3. Logical Operations" << endl;
    cout << "Enter your choice: ";
    cin >> choice;

    switch (choice) {
    case 1: {
        int num1, num2;
        cout << "\nEnter first number: ";
        cin >> num1;
        cout << "Enter second number: ";
        cin >> num2;
        sum(num1, num2);
        break;
    }

    case 2: {
        int A, B;
        cout << "\nEnter value of A: ";
        cin >> A;
        cout << "Enter value of B: ";
        cin >> B;
        cout << "Greater value: " << greaterNum(A, B) << endl;
        break;
    }

    case 3: {
        bool x, y;
        cout << "\nEnter first value (1 or 0): ";
        cin >> x;
        cout << "Enter second value (1 or 0): ";
        cin >> y;
        logicalOperation(x, y);
        break;
    }

    default:
        cout << "Invalid choice!" << endl;
```

```
51      }
52
53      return 0;
54  }
55
56
57  void sum(int a, int b) {
58      cout << "Sum: " << a + b << endl;
59  }
60
61  int greaterNum(int A, int B) {
62      if (A > B)
63          return A;
64      else
65          return B;
66  }
67
68  bool logicalOperation(bool a, bool b) {
69      cout << "AND: " << (a && b) << endl;
70      cout << "OR: " << (a || b) << endl;
71      cout << "NOT A: " << (!a) << endl;
72      cout << "NOT B: " << (!b) << endl;
73
74      return true;
75  }
```

```
MENU
1. Sum of Two Numbers
2. Greater Number
3. Logical Operations
Enter your choice: 1

Enter first number: 2
Enter second number: 3
Sum: 5

--------------------------------
Process exited after 6.19 seconds with return value 0
Press any key to continue . . .
```

Del Mundo.cpp

```cpp
#include <iostream>
using namespace std;

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    const bool validateTriangle();
};

Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

const bool Triangle::validateTriangle() {
    return (totalAngle <= 180);
}

int main() {
    Triangle set1(40, 30, 110);

    if (set1.validateTriangle()) {
        cout << "The shape is a valid triangle.";
    }
    else {
        cout << "The shape is NOT a valid triangle.";
    }

    return 0;
}
```

```
The shape is a valid triangle.
--------------------------------
Process exited after 0.05431 seconds with return value 0
Press any key to continue . . .
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Del Mundo.cpp

```cpp
#include <iostream>
#include <cmath>
using namespace std;

void swapNumbers(int &, int &);
double kelvinToFahrenheit(double);
double distance(double, double, double, double);

int main()
{
    int choice;

    cout << "MENU" << endl;
    cout << "1. Swap Two Numbers" << endl;
    cout << "2. Kelvin to Fahrenheit" << endl;
    cout << "3. Distance Between Two Points" << endl;
    cout << "Enter your choice: ";
    cin >> choice;

    if (choice == 1)
    {
        int num1, num2;

        cout << "Enter first number: ";
        cin >> num1;

        cout << "Enter second number: ";
        cin >> num2;

        cout << "Before Swapping:" << endl;
        cout << "First Number = " << num1 << endl;
        cout << "Second Number = " << num2 << endl;

        swapNumbers(num1, num2);

        cout << "After Swapping:" << endl;
        cout << "First Number = " << num1 << endl;
        cout << "Second Number = " << num2 << endl;
    }
    else if (choice == 2)
    {
        double kelvin, fahrenheit;

        cout << "Enter temperature in Kelvin: ";
        cin >> kelvin;

        fahrenheit = kelvinToFahrenheit(kelvin);

        cout << "Temperature in Fahrenheit = " << fahrenheit << endl;
    }
```

```
51      else if (choice == 3)
52      {
53          double x1, y1, x2, y2, result;
54
55          cout << "Enter x1: ";
56          cin >> x1;
57
58          cout << "Enter y1: ";
59          cin >> y1;
60
61          cout << "Enter x2: ";
62          cin >> x2;
63
64          cout << "Enter y2: ";
65          cin >> y2;
66
67          result = distance(x1, y1, x2, y2);
68
69          cout << "Distance = " << result << endl;
70      }
71      else
72      {
73          cout << "Invalid choice!" << endl;
74      }
75
76      return 0;
77  }
78
79
80  void swapNumbers(int &a, int &b)
81  {
82      int temp;
83
84      temp = a;
85      a = b;
86      b = temp;
87  }
88
89  double kelvinToFahrenheit(double k)
90  {
91      return (k - 273.15) * 9 / 5 + 32;
92  }
93
94  double distance(double x1, double y1, double x2, double y2)
95  {
96      return sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
97  }
```

```
MENU
1. Swap Two Numbers
2. Kelvin to Fahrenheit
3. Distance Between Two Points
Enter your choice: 1
Enter first number: 3
Enter second number: 2
Before Swapping:
First Number = 3
Second Number = 2
After Swapping:
First Number = 2
Second Number = 3

--------------------------------
Process exited after 12.42 seconds with return value 0
Press any key to continue . . .
```

Sorry sir kung puro codings lang po. Ngayun lang po kase nagawa since wala po akong laptop huhu
Bawi ako next lab sir. Thank you!

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Chat gpt and grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

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Purpose of Use: **Code assistance and explanation of concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **I use it to explain the codes section by sections**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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